Factorisation Method for the Sech-Squared Potential

Source: Cambridge Quantum Mechanics, Example Sheet 1, Question 8; accompanies David Tong's notes.

Sketch the potential

$$U(x) = -\frac{\hbar^2}{m}\operatorname{sech}^2 x\,. \tag{1.1}$$

Show that the time-independent Schrödinger equation for a particle in this potential can be written

$$\hat{A}^\dagger \hat{A}\,\chi = (\mathcal{E}+1)\,\chi\,, \tag{1.2}$$

where $\mathcal{E} = 2mE/\hbar^2$ and

$$\hat{A} = \frac{d}{dx} + \tanh x\,, \qquad \hat{A}^\dagger = -\frac{d}{dx} + \tanh x\,. \tag{1.3}$$

Show, by integrating by parts, that for any normalised wavefunction χ,

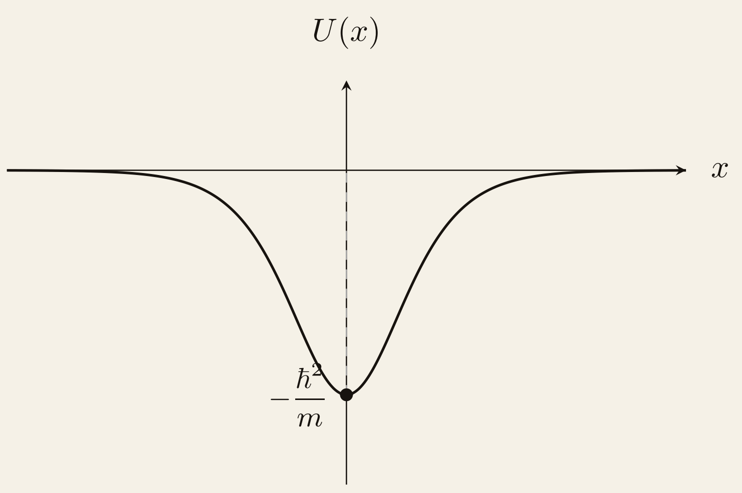
$$\int_{-\infty}^{\infty} \chi^* \hat{A}^\dagger \hat{A}\,\chi\,dx \;=\; \int_{-\infty}^{\infty} (\hat{A}\chi)^*(\hat{A}\chi)\,dx\,, \tag{1.4}$$

and deduce that the eigenvalues of $\hat{A}^\dagger\hat{A}$ are non-negative. Hence show that the ground state (with lowest energy) has $\mathcal{E} \geq -1$. Show that a wavefunction $\chi_0(x)$ is an energy eigenstate with $\mathcal{E} = -1$ iff

$$\frac{d\chi_0}{dx} + \tanh x\,\chi_0 = 0\,. \tag{1.5}$$

Find and sketch $\chi_0(x)$.

Solution.



The sech-squared potential \eqref{eq:sech-pot

The Hamiltonian can be written as

$$\begin{aligned} H &= -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} - \frac{\hbar^2}{m}\operatorname{sech}^2 x \\ &= \frac{\hbar^2}{2m}\left(-\frac{d^2}{dx^2} - 2 + \tanh^2 x\right) \\ &= \frac{\hbar^2}{2m}\left(\frac{d}{dx} + \tanh x\right)\left(-\frac{d}{dx} + \tanh x\right) - \frac{\hbar^2}{m}. \end{aligned} \tag{1.6}$$

Thus, we have

$$\hat{A}^\dagger \hat{A}\chi = (\mathcal{E}+1)\chi. \tag{1.7}$$

Rather trivially following from integrating by part, we find

$$\int dx \chi^* A^\dagger A \chi = \int dx (A\chi)^*(A\chi), \tag{1.8}$$

justifying that $A^\dagger$ is indeed the Hermitian conjugate to A. Given an eigenstate of $A^\dagger A$, $|a\rangle$, with eigenvalue a. We have

$$\langle a|A^\dagger A|a\rangle = a\langle a|a\rangle = ||A|a\rangle||^2 \geq 0 \implies a \geq 0, \tag{1.9}$$

which implies that $\mathcal{E}+1 \geq 0$. For the ground state with energy $\mathcal{E} = -1$, we have

$$\hat{A}^\dagger \hat{A}\chi_0 = 0, \tag{1.10}$$

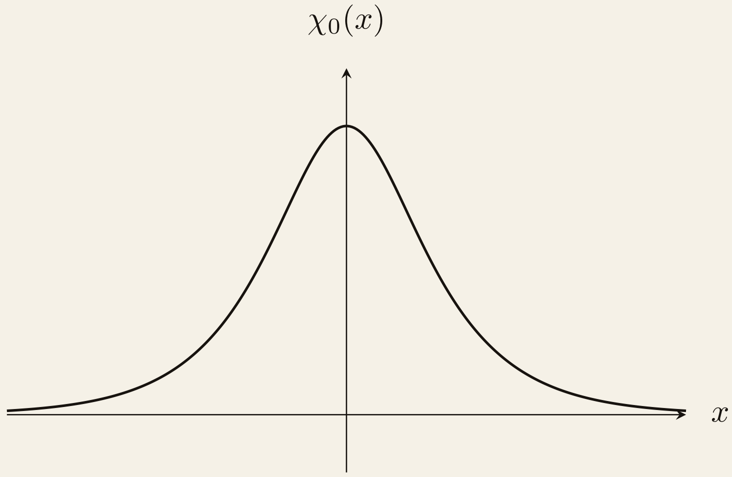
Taking the inner product of the above with χ_0, we find

$$\langle \chi_0|A^\dagger A|\chi_0\rangle = ||A|\chi_0\rangle||^2 = 0 \implies A|\chi_0\rangle = 0. \tag{1.11}$$

Solving the above, we find

$$\chi_0(x) = \frac{N}{\cosh x}, \tag{1.12}$$

where N is a normalisation constant.



Ground state wavefunction $\chi_0(x) = N \operatorname{sech} x$, peaked at the origin and decaying exponentially as $x \to \pm\infty$.